

INPUTS

SCORING — AUTOMATED PROXIES

COMPOSITE

DECISION

Candidate dataset D

with documentation /
datasheet

Modality m

text · tabular · image

Selects proxy

Safety sub-rubric (Table I)

Content-origin (CO)
poisoning susceptibility · adversarial provenance ·
factual decay · harm-content density ●
Physical-safety (PS)
physical-harm enablement · edge-case coverage

computes scores

Automated proxies — modality-specific

Detoxify → text harm-content ●
confident-learning estimator → tabular integrity ●
image content proxy → extension point (not yet validated)
domain half-life constants → factual decay

Re-assess on dataset repurposing — planned extension (Sec. VII)

Safety composite S(D)

unweighted mean of
applicable CO + PS
sub-dimensions.
N/A sub-dims removed from
denominator (not zeroed)

Threshold rule

deployer-set, per score
e.g. $S \leq 0.2$ (high-stakes)
physical-harm flagged for
review — not auto-excluded

Store score

S(D) + dataset hash
with metadata

● = Proxy empirically validated in Sec. VI (text harm-content density; tabular label-integrity)